

WASP-50b

R-1063

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_161024 · created 2026-07-30T12:30:30+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457685.752 +/- 0.035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.02 +/- 1.78 [%]
Orbital Inclination (inc)	88.59 +/- 2.9
Airmass coefficient 1 (a1)	0.91 +/- 0.47
Airmass coefficient 2 (a2)	-0.01 +/- 0.17
Scatter in the residuals of the lightcurve fit is	50.75 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	6.75
Transit Duration (day)	0.087 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.051 vs 0.1404 (deviation 0.61σ)
Duration fit vs published	0.087 d vs 0.0754 d (deviation 0.71σ)
Mid-time O–C	-2.5 min
Depth SNR	3.1
Residual scatter	50.75 %

Light curve

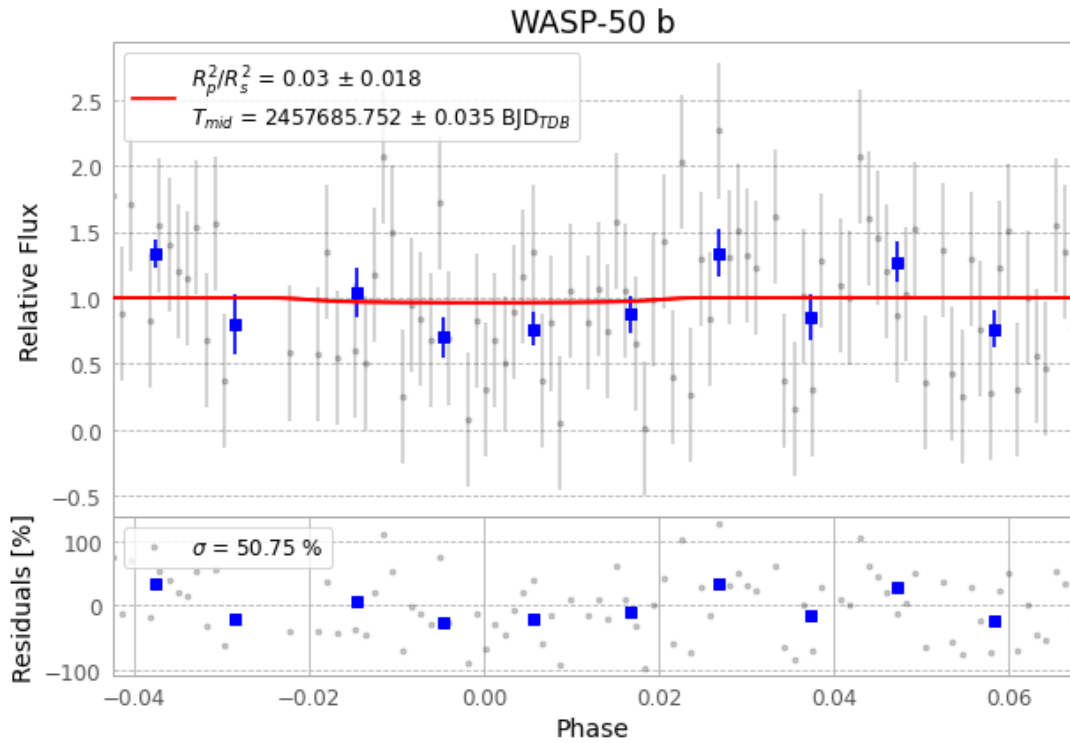


Figure 1 — FinalLightCurve_WASP-50 b_2016-10-23.png (SHA-256 dc1f90c28d5831ad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [61, 103], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d24ce9f54cd34fa5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-50161024035412.FITS → WASP-50161024090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-161024.log (b4d670585fd8...), FinalLightCurve_WASP-50 b_2016-10-23.png (dc1f90c28d58...), FinalLightCurve_WASP-50 b_2016-10-23.csv (3e3f0cf377b1...)

Compute resources

Wall time 501.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 12:30Z.